

Network of Democratic Economic Cells (NDEC v5)

From the National Base to the Planetary Horizon — Patch v4.4.3: Use-Property & Defense of the Commons

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Note: V4.4.3 was renamed V5 due to a new theoretical core being integrated.

Application Note for Patch v4.4.3

The patch file specifies a source version of 4.4.2 and a target version of 4.4.3. As v4.4.2 is unavailable, the patch has been directly applied over the v4.4.1 base. No content from v4.4.1 has been deleted. The numbering conflict has been resolved: the *Ecological Loop* remains Layer 12, and the defense mechanisms introduced by the patch are appended as separate layers (Guard of the Commons under internal security, followed by Layers 15 Territorial Defense, 16 Immunity, and 17 Verification). These additions fall under Part II (aspirational horizon, not deliverable within 20 years) under a cautious framework: consent, civil control, interposition over imposition, last resort, and mandatory prior human validation.

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1. Executive Summary

The NDEC model proposes a radical yet pragmatic transformation of the Swiss economic system. Version v4.4.3 preserves all the achievements of the core base (v4.3.2 and Layers 13-14 from v4.4.1): blockchain-based transparency, cooperative governance, democratic control, common infrastructure, an ecological loop, and the Decarbonization Fund. Over this, it applies Patch v4.4.3, which drives a major paradigm shift and formalizes the defense of the commons.

Patch v4.4.3 hinges on four pillars: (1) capitalist private property yields to use-property; (2) traditional redistributive taxation is replaced by a Commons Use Fee; (3) the model transitions from a mere fiscal transition tool to a structure of direct social production; (4) the defense of the commons is rigorously structured (Guard of the Commons, territorial defense, immunity, verification).

Key Results — Baseline Scenario (Year 20)

Projections for the Baseline Scenario in Part I demonstrate exceptional macroeconomic and ecological performance without monetary creation, strictly adhering to gain allocation constraints.

+180.4% GDP Index Base 2026 = 100 (+180.4 pts)	72 / 100 Gross National Happiness Increase of +14 points	0.185 Gini Coefficient Strong reduction (-0.135)
28.5 hrs Working Hours / Week Liberated time (-12.5 hrs)	-68% Net Emissions 2050 Net Zero Trajectory (LCI)	19.7% Sovereign Fund Surplus Expressed as % of GDP

Complementary Indicator	Value at Y20	Impact / Justification
Shared Commons Networks	3 networks	Rail, energy, and digital networks open and rent-free.
Final Energy (Index Y20)	77.4	22.6% reduction in final energy consumption.
Required Overbuild (Layer 14)	1.25x	Optimized vs ~2x required without a wide interconnected grid.

2. Evolutions from v4.3.2 to v4.4.3

The v4.4.x version does not invalidate any corrections from version v4.3.2 but expands upon them. The baseline (Layers 0 to 12) and the eight digital consistency checks remain unchanged. The model anchors itself in an incremental logic building on existing precedents (SBB, Swissgrid, civil service, EU citizen energy communities) while introducing foundational tenets from Karl Marx (property), Elinor Ostrom (commons and graduated sanctions), and radical subsidiarity.

Element (Base)	Evolution with Patch v4.4.3	Justification / Impact
Reallocation γ for non-automatable sector	Layer 13: qualifying citizen service (renovation, energy, care).	Provides a concrete operator to meet the 3x to 5x pace required for a 100% renewable system.
Decarbonization Fund (δ)	Layer 13: low-tech thermal storage (sand) + local resources.	Removes dependency on critical minerals; seasonal storage; lowers final consumption.
National Commons (Layer 11)	Layer 14: federated energy cells, energy subsidiarity.	Smooths intermittency across geographical space, reducing overbuild from $\sim 2x \rightarrow 1.25x$.
International Grid Limitations	Part II: Earth Federation Project, sequenced after baseline maturity.	Addresses the political blind spot as a distinct horizon separated from deliverable layers.
Capitalist Private Property & Redistributive Taxation	Institutionalized Use-Property and Commons Use Fee (Layer 2).	Assets are legally tied to their actual use. Replaces post-hoc redistribution with source-level fees.
Financial Asset Register (Layer 1)	Usage Cadastre (energy, land, digital).	Auditing focuses on the actual use of resources, aligning with use-property.
Collective Security (§12, horizon)	Guard of the Commons + Layers 15-17 (defense, immunity, verification).	Graduated defense of the commons: enclosure prevention, automated immunity, prior human validation.

3. Paradigm Shift (Patch v4.4.3)

Core Purpose: The model is no longer framed as a mere fiscal transition tool, but as a **structure of direct social production**: economic coordination occurs through the shared management of the commons rather than capital accumulation followed by late-stage redistribution.

3.1 From Private Property to Use-Property

The patch dismantles private property as defined by capitalism (capital titles conveying the right to exclude and extract rent). It substitutes it with **use-property**: an asset is constitutionally tied to whoever actively uses it in accordance with the rules of the common, rather than to a detachable title. Inspired by Marx read through Ostrom, this principle does not prohibit owning what you use (housing, tools, farms), but revokes the ability to capture rent on an unused asset, which deprives others of its use.

3.2 From Redistributive Taxation to the Commons Use Fee

The traditional system of redistributive taxation (Piketty / Zucman / Saez) is entirely replaced by a **Commons Use Fee**. Instead of levying taxes *ex-post* on income and wealth to redistribute them later, the model bills the use of common resources (energy, land, data, registry capacity) directly at the source. The contribution becomes a condition for accessing the common, falling in line with the Ostromian logic of access-dues. Layer 0 (inequality diagnosis) remains valid as an initial analytical reference, but the operational tool shifts to the use fee.

4. The Complete Layer Stack (Base + Patch)

The global framework integrates the historical baseline layers reinterpreted through Patch v4.4.3, supplemented by ecological transition innovations (13-14) and security systems (15-17).

#	Element / Layer	Institutional Function & Mechanism
0	Diagnosis	Assessment of inequalities (analytical point of departure, no longer an action mechanism).
1	Usage Register	Blockchain-based cadastre tracking energy, land, digital, and carbon usage (Patch v4.4.3).
2	Usage Fee	Replaces redistributive taxation: source-level billing for the use of the commons (Patch v4.4.3).
3	Capitalization	Citizen sovereign fund investing directly into the real economy.
4a	Production	Worker cooperatives.
4b	Consumption	Consumer cooperatives and democratic supply-side planning.
5	Health / Time	Solidarity-based insurance and implementation of a reduced workweek.
6	Coordination	Federated cells; free market utilized strictly as a transmission signal.
7	Governance	Swiss model founded on strict subsidiarity, from municipalities to the Confederation.
8	Knowledge	Official training framework (SERI / OrTra) with certified retraining paths.
9	Propagation	0% interest investments enabling the expansion and seeding of cells internationally.
10	Time	Labor indexation: required working hours decrease proportionally to automation.
11	National Commons	Open-access rail, energy, and digital infrastructure to purge rent extraction.
12	Ecological Loop	Strict carbon budget + Decarbonization Fund (δ) — LCI 2050 Net Zero Trajectory.
13	Citizen Mobilization	Citizen service + low-tech thermal storage + structural relocalization (§ 5).
14	Energy Cells	Nested energy sharing and networked energy subsidiarity (§ 6).
I.S.	Guard of the Commons	Internal security: enclosure prevention, technical auditing, administrative interposition.
15	Territorial Defense	Subsidiarity-based, decentralized external defense, funded by the commons dividend.

#	Element / Layer	Institutional Function & Mechanism
16	Immunity	Broadcast protocol: automated, reversible isolation of non-compliant cells.
17	Verification	Decentralized media consensus (human-in-the-loop) to validate real-world aggression.

4.1 Gain-Sharing Mechanism (Reminder)

For any given year t , the annual automation rate $\Delta P(t) = r$ (estimated between 2% and 6%) is rigorously distributed according to the following allocation constraint:

$$\text{Allocation Constraint: } \alpha + \beta + \gamma + \delta (+ \epsilon) = 1$$

The baseline coefficients for the operational core are defined as follows:

- $\alpha = 45\%$ → Funds workweek reduction ($\alpha \times \Delta P(t)$).
- $\beta = 35\%$ → Feeds the Citizen Sovereign Fund ($\beta \times \Delta P(t)$).
- $\gamma = 10\%$ → Allocated to sector reallocation and training / Layer 13 ($\gamma \times \Delta P(t)$).
- $\delta = 10\%$ → Directed to the Decarbonization Fund / Layer 12 ($\delta \times \Delta P(t)$).
- ϵ (Federal Contribution / Part II): Conditional, activated only if the Sovereign Fund surplus exceeds 2% of GDP.

Golden Rule: 100% of automation gains are allocated, with absolutely zero monetary creation.

Democratic Governance Note

The annual arbitration and adjustment of macroeconomic coefficients ($\alpha, \beta, \gamma, \delta, \epsilon$) are not managed by frozen algorithms or technocratic bodies. Budget allocation is sovereignly debated and ratified by a **Citizen Assembly**, ensuring that production priorities and work hours reduction continuously align with the population's expressed needs.

5. Layer 13 - Citizen Mobilization for the Transition

The deployment pace required to substitute fossil fuels with renewable energy sources is 3 to 5 times faster than the current rate. The conventional labor market cannot supply this workforce without problematic delays. Layer 13 deploys three major levers to accelerate this transition while lowering final energy consumption without eroding societal well-being.

5.1 Transition Citizen Service

The principle consists of repurposing a consensual and deeply rooted Swiss institution (compulsory military/civil service) rather than engineering unprecedented coercion. Able-bodied young adults perform a qualifying service, supervised by professionals, channeled into critical transition work: energy retrofitting of buildings, deployment of renewable infrastructures and heat pumps, ecological restoration, and care sector support.

- **Link with Layer 8 (Knowledge):** The program culminates in official certifications or federal diplomas; mobilization and vocational training merge into a single mechanism.

- **Link with the non-automatable sector:** The service feeds workforce into ecology, healthcare, and education, directly nourishing Gross National Happiness (GNH).

Imperative Safeguards to Integrate:

1. **Individual Liberty:** The obligation remains a constraint; it is justified only if the service remains brief, highly qualifying, and provides a real choice of tracks.
2. **Productivity:** An untrained conscript is inherently less efficient than a professional artisan. The service must complement qualified trades rather than replace them, requiring dense professional oversight.
3. **Accounting:** Opportunity costs must be strictly accounted for as a workforce transfer (not free labor) within the consolidated balance sheet.

5.2 Low-Tech Thermal Storage — Sand Batteries

A fundamental technical distinction must be made: **we do not store electricity, we store heat**. Surplus renewable electricity is used to heat sand to temperatures of 500-600 °C. This sand then feeds district heating networks over days or months.

What sand storage solves	What must be lucidly acknowledged
Mineral Independence: Zero reliance on lithium, cobalt, nickel, or rare earths. Uses only sand, steel, and a heating element. A local, abundant, and impact-neutral resource.	Mediocre Electrical Efficiency: High energy losses if converting heat back into electricity. Unbeatable for heating, but unsuited for mobility.
Seasonal Storage: Capable of transferring summer energy surpluses into winter, something classic electrochemical storage cannot achieve at scale.	Heavy Infrastructure: Presupposes the existence or dense rollout of district heating networks, which represents a massive infrastructure project in its own right.
Peak Management: Efficiently absorbs sharp peaks of renewable overproduction instead of relying on curtailment.	Complementarity: It is an indispensable complement to, and not a replacement for, electrochemical storage (which remains reserved for mobility and electronics).

5.3 Retrofitting, Relocalisation, and Energy Balance

In countries with temperate or cold climates, space heating accounts for nearly 45% of final energy consumption. Deep thermal retrofits of building envelopes cut heat demand by 50 to 80%, while heat pumps (HP) with a COP of 3 to 4 drastically divide the required primary energy.

Transition Domain	Measured Effect on Final Consumption
Building Heating (Retrofits + HP)	↓↓↓ Dominant and highest priority lever.
Freight Transport (Relocalisation)	↓ Net reduction in physical trade flows.
Seasonal Storage (Sand Batteries)	Neutral Enables securing a 100% renewable thermal grid.

Transition Domain	Measured Effect on Final Consumption
Specific Electricity (Global Electrification)	→ or ↑ Flat to slight increase in net grid demand.
Grey Energy of Transition (Initial Construction)	↑ Temporary surge, amortized down the line.
Net Balance (Index Y20 = 77.4)	Overall 22.6% reduction in final consumption, at constant well-being.

6. Layer 14 - Federated Energy Cells

The remaining blind spot of the transition concerns the specific intermittency of electricity. Layer 14 responds by rolling out grid-based sharing, acting in perfect synergy with storage solutions.

The model links two physical dimensions: **time** (shifting energy from one moment to another via local storage) and **space** (shifting energy from one geographic node to another via a meshed grid to smooth out instantaneous production variations).

Scale	What spatial sharing resolves	What it cannot resolve on its own
Neighborhood	Micro-diversity of demand, line loss reduction, optimization of local self-consumption.	Day/night cycles, highly localized weather events (zonal production remains correlated).
National	Smoothing of regional microclimates and weather discrepancies.	Day/night cycles (Switzerland's longitudinal expanse is too narrow).
Continental	Strong weather compensation, time-zone shift exploitation, and wind/solar anti-correlation.	Deep, persistent winter windless/sunless low-production spells (<i>Dunkelflaute</i>).
Global	Complete resolution of the day/night cycle (it is always daytime on one side of the planet).	Physically robust, but reopens the immense challenge of global political governance (see Part II).

6.3 Governing Principle: Energy Subsidiarity

The grid operates on a fractal, nested architecture: local self-sufficiency is maximized in the first instance. Higher tiers only step in to balance the residual energy that lower tiers cannot absorb. Territorial autonomy is the unyielding rule, global interconnection is the safety net (and never the reverse). Cells do not depend on the macro-grid; they connect to it for insurance purposes.

Impact on Overbuild: Thanks to this broad network and Layer 14's meshing, the *overbuild* index (the required over-sizing of production infrastructure) collapses from **~2x (for an isolated island system) to just 1.25x**, massively saving grey energy and critical minerals.

Technical Integration: Layer 1 (Blockchain) transparently traces energy flows and peer-to-peer reciprocity credits between cells. Real-world regulatory precedents anchor this approach: Self-Consumption Communities (RCP/ZEV) in Switzerland and Citizen Energy Communities championed by the European Union.

Double Trajectory Note

The long-term viability of the model depends on its outward spread. A symmetrical effort of cultural animation and diplomacy is required: internally (rolling out pilot portfolios across cantons, local carbon accounting) and internationally (citizen and academic diplomacy to standardize cell-based accounting and prime Part II).

PART II — EARTH FEDERATION PROJECT (ASPIRATIONAL HORIZON)

Status: Multi-generational, aspirational project; not deliverable within a 20-year horizon. Its activation remains strictly conditional on the full maturity of Part I (defined by the propagation of the cell network to a critical mass of nations).

8. Status, Sequencing, and Nature of the Problem

The Earth Federation differs fundamentally in nature from Layers 13 and 14. The latter resolved technical constraints through engineering. The Earth Federation purports to bridge the **political blind spot**: global coordination. However, sovereignty cannot be erased by mere design tricks. The two projects are therefore strictly sequenced: each phase must prove its absolute robustness before the next can be envisioned.

10. Equalization via Commons Dividend

To remain realistic and politically tractable, global equalization is not organized as a direct wealth transfer from rich to poor continents, but is backed by a **dividend from planetary commons** themselves. The atmosphere (via a unified carbon registry), the international energy backbone, orbital space, oceans, and critical mineral deposits are defined as global commons.

Usage fees levied at the source on these resources supply a Sovereign Fund transposed to a planetary scale. Redistribution occurs out of these centralized revenues: the paradigm is no longer "the rich pay for the poor," but "the global common yields its dividend and distributes it according to need."

11. Financing: 0% Investment → Commons → Dividend

The global financial engine replicates Layer 9's propagation mechanism, inspired by major historical cooperative federations (Migros, Coop, Raiffeisen, Mondragon) where member surpluses fund umbrella structures.

- **Investment:** Surpluses from national economic cells are funneled, via 0% interest loans, into building global infrastructures (continental interconnectors, registries, environmental restoration).
- **Equalization:** Once operational, these common infrastructures generate access fees. This commons dividend then funds structural equalization.

Enforcement Without Armies (Access-Dues): Drawing from Ostrom's work, global contributions are not enforced via invasive customs or military coercion, but take the form of an unavoidable access condition to the commons network. To trade, use energy infrastructure, or utilize the carbon registry, a cell must be up-to-date with its access-dues. Compliance is not enforced by arms, but by technical disconnection.

14. Collective Security: From Non-Violent Leverage to Interposition Force

This tier represents the most sensitive and strictly regulated element of Part II. Conceived strictly as a last resort, it entirely excludes a permanent centralized army and relies on two graduated instruments:

Instrument	Nature of Dispositive & Mechanism	Recourse
Non-violent Leverage	Exclusion from accessing the commons (carbon register, unified market, energy interconnectors). No wealth is extracted; the non-compliant cell is technically unplugged.	First Recourse Soft and economic
Consensual Interposition Force	Physical deployment between belligerents to freeze hostilities and prevent renewed violence. Strict Peacekeeping logic, never assault troops.	Last Resort Heavy and exceptional

Absolute Red Lines and Security Architecture:

- **Interposition, Not Imposition:** The federal force does not deploy *against* a side to subdue or occupy it. It does not govern and levies no taxes.
- **Prior Consent:** Entry is strictly conditional on invitation or a consented mandate; entering the territory of a sovereign state by force without consent is absolutely prohibited.
- **The Execution Dilemma:** Lacking direct coercive power, the global registry risks becoming a mere scoreboard; conversely, if endowed with enforcing force, it risks drifting into an authoritarian World-State—the exact dangerous structure the NDEC model fights against. An Ostromian confederal approach with strictly enumerated powers serves as the firebreak.

Appendix A — Mathematical Formulas

F1 — Extended Macroeconomic Allocation Constraint:

$$\alpha + \beta + \gamma + \delta (+ \epsilon) = 1$$

Where $\epsilon \geq 0$ represents the federal contribution (Part II), capped and strictly zero if the Sovereign Fund surplus is $\leq 21\%$ of GDP.

F2 — Renowated Building Heat Demand & HP Consumption:

$$Q_{heat}(t) = Q_{heat}(0) \times (1 - \tau_{retro})$$

$$E_{elec} = Q_{heat} / COP$$

Target Parameters: Reduction rate $\tau_{retro} \approx 0.5 \text{ ext{ to } } 0.8$ | Coefficient of Performance $ext{COP} \approx 3 \text{ ext{ to } } 4$.

F3 — Seasonal Thermal Storage (Sand Batteries):

$$S_{th}(t) = S_{th}(t-1) + \eta_{charge} \times E_{surplus}(t) - Q_{restored}(t) - losses(t)$$

Optimized thermal-to-thermal cycle for absorbing immediate renewable generation peaks.

F4 — Cell Energy Balance (Layer 14):

$$SelfConso = \min(ext{Prod}_{local}, ext{Conso}_{local})$$

$$Surplus_{shared}(t) = \sum \max(ext{Prod} - ext{Conso}, 0) \rightarrow ext{deficit cells}$$

Target Infrastructure Overbuild (v4.4 target): $1.25\times$ (versus $\sim 2\times$ without a meshed grid).

F5 — Cascaded Federal Financing (Part II):

$$Invest_{commons}(t) = \epsilon \times \Delta P(t) \times ext{GDP}(t)$$

$$Dividend_{commons}(t) = ext{carbon_price} \cdot E_{taxed} + ext{mineral_royalties} + ext{user_fees}$$

$$Equalization(t) = ext{allocation}(ext{Dividend_commons})$$

Appendix B — Glossary (New Entries via Patch v4.4.3)

Term	Definition and Systemic Role
Use-Property	A legal framework binding an asset to its active user under the terms of a common charter. Abolishes capital titles that allow passive rent extraction (Marx read through Ostrom).
Commons Use Fee	A mechanism billing the usage of common resources directly at the source, fully replacing old post-hoc redistributive taxation systems (Layer 2).
Guard of the Commons	An internal administrative and civil body responsible for auditing the registry (Layer 1), alerting assemblies, and issuing administrative interpositions (access lockouts) to counter enclosure attempts.

Term	Definition and Systemic Role
Immunity (Broadcast)	An automated, reversible technical protocol that isolates any economic cell violating established common rules. An operationalization of Ostrom's graduated sanctions.
Verification	A decentralized media consensus process mandating prior human validation (human-in-the-loop) to certify real-world physical aggression or breaches of the commons before a collective response can be triggered.
Interposition Force	An international peacekeeping force consisting of a slim professional core backed by a citizen volunteer reserve. Deploys strictly between belligerents to freeze violence.
Citizen Assembly	A sovereign deliberative organ responsible for debating and adjusting macro coefficients (α , β , γ , δ , ϵ) annually, removing technocratic or automated rigidity.

Appendix C — Sources and Bibliographical References

- **Ostrom, E. (1990).** *Governing the Commons*. — General theory of the commons, access-dues, nested institutional architectures, and graduated sanctions.
- **Marx, K.** *Das Kapital / Manifesto*. — Radical critique and framework for dismantling capitalist private property, operationalized here through use-property.
- **Piketty, T. (2013).** *Capital in the Twenty-First Century*. — Historical empirical mapping of capital concentration, serving as the base for the inequality diagnosis (Layer 0).
- **Saez, E. & Zucman, G. (2019).** *The Triumph of Injustice*. — Empirical analyses of contemporary taxation systems and redistributive dynamics.
- **Polar Night Energy (2022).** — First commercial-scale industrial deployment of high-temperature sand thermal storage in Finland (Technical reference for Layer 13).
- **Cooperative Models (Mondragon, Migros, Raiffeisen).** — Frameworks for confederal architectures funding collective development and solidarity mechanisms from member surpluses.
- **Swiss RCP/ZEV Regulations.** — Legal framework for self-consumption communities, serving as an empirical blueprint for Layer 14's energy cells.